SC-KV-05058

Kria KV260 Vision AI Starter Kit Carrier Card

AMD

AECG
2100 Logic Drive
San Jose,CA 95124

PAGE: **01 - Title Page**

PAGE NUMBER: **1** OF **18**

DOC NO: **038-05058-03**

DATE: **Wednesday, May 03, 2023** REV: **A1**

SKU: **SKIT**

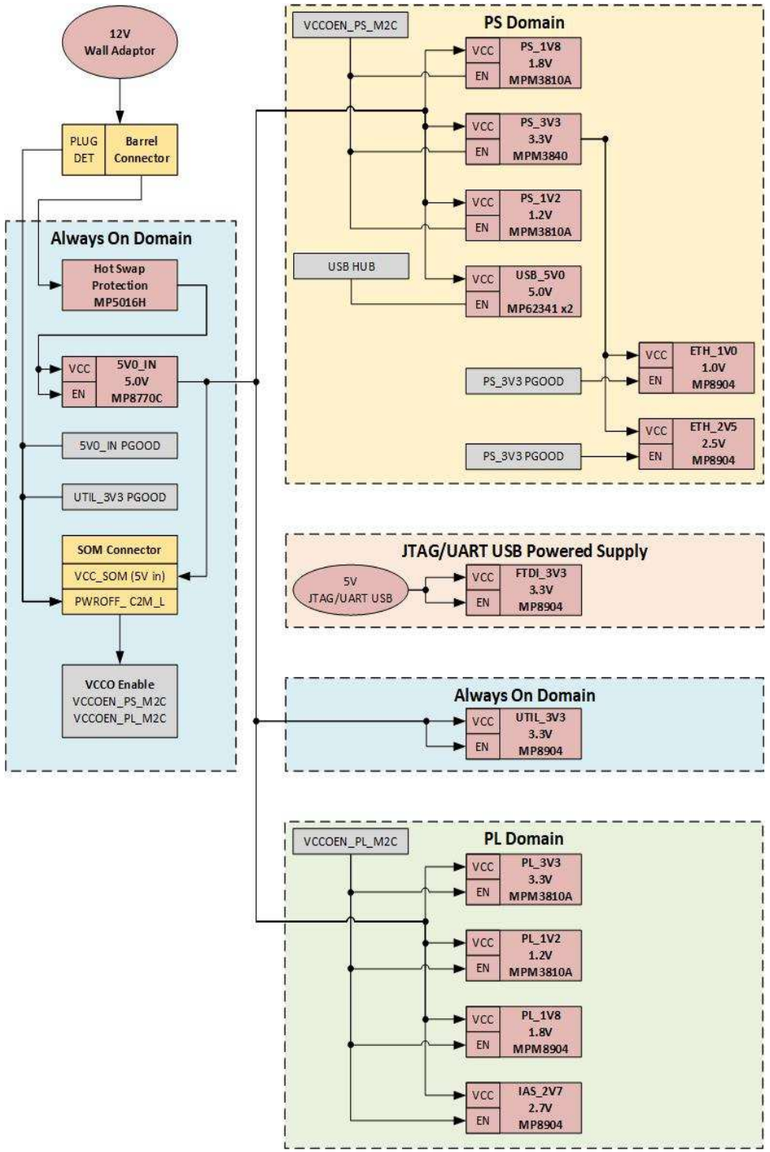
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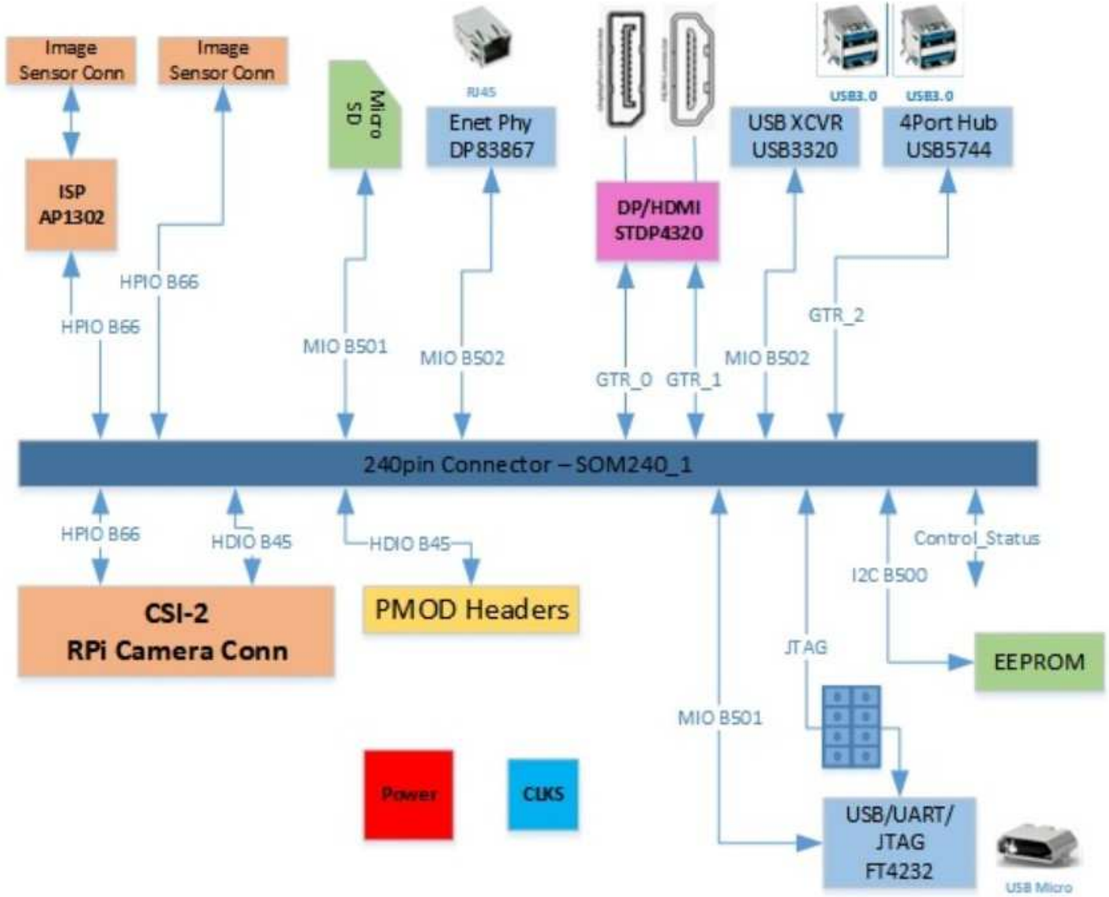
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TITLE:**SC-KV-05058**

POWER BLOCK DIAGRAM



PERIPHERAL BLOCK DIAGRAM



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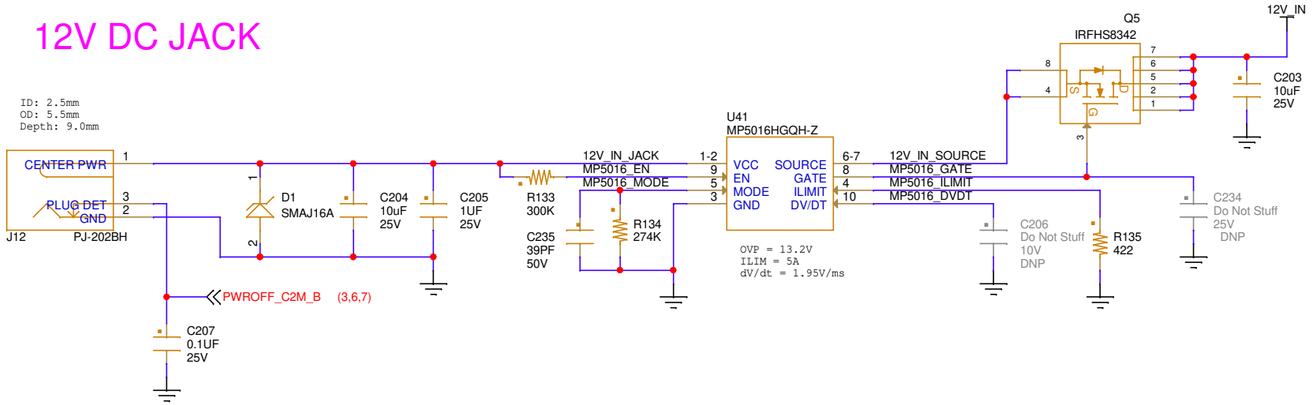
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PAGE: **02 - Block Diagram**	
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DOC NO: **038-05058-03**	
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SKU: **SKIT**	

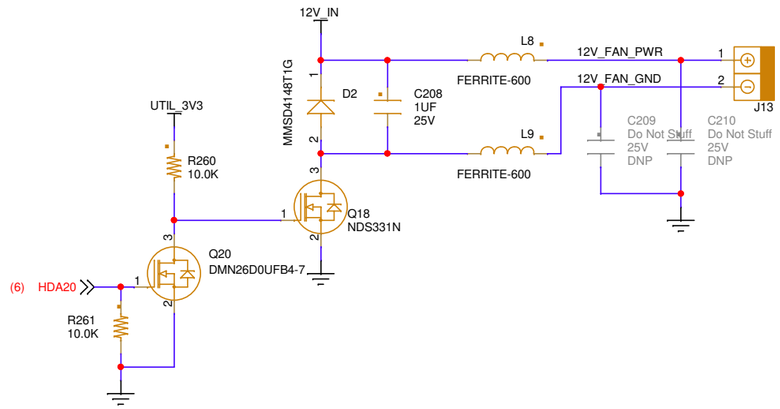
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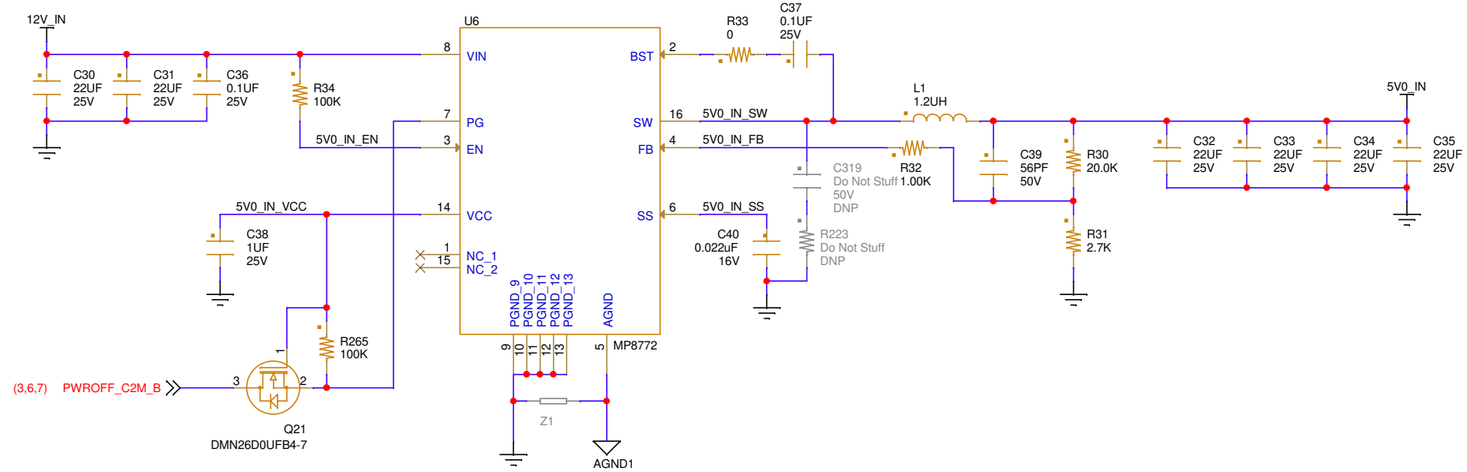
12V DC JACK



12V FAN HEADER



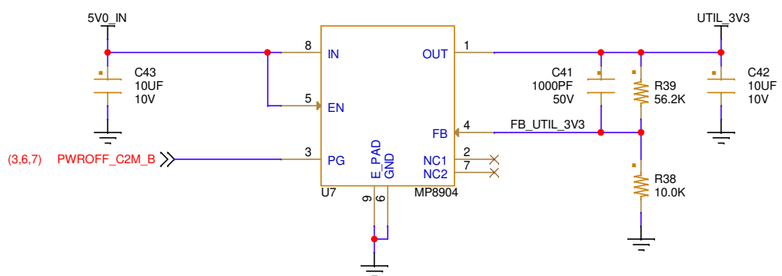
5V0_IN ALWAYS ON REGULATOR



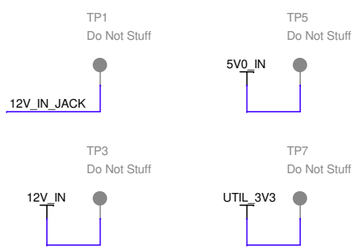
SOM_5V0 CURRENT MONITOR

INA260 ADDR: 0x40

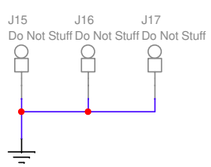
UTIL_3V3 ALWAYS ON REGULATOR



POWER TEST POINTS



GROUND LOOPS



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PAGE:	**03 - Always On Pwr, Fan Header**
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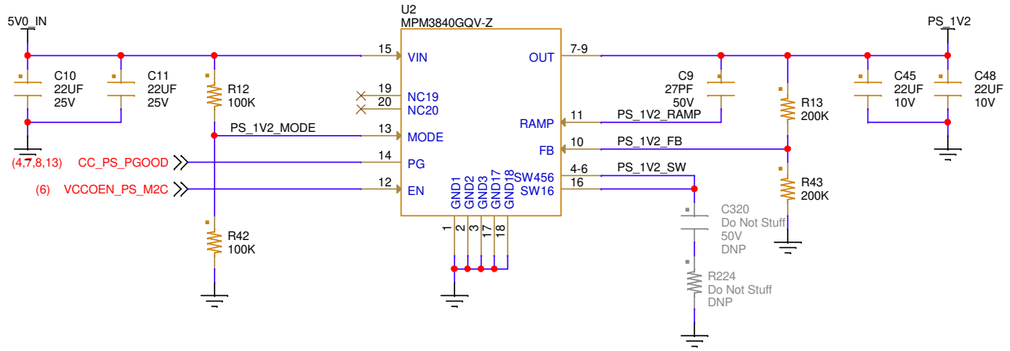
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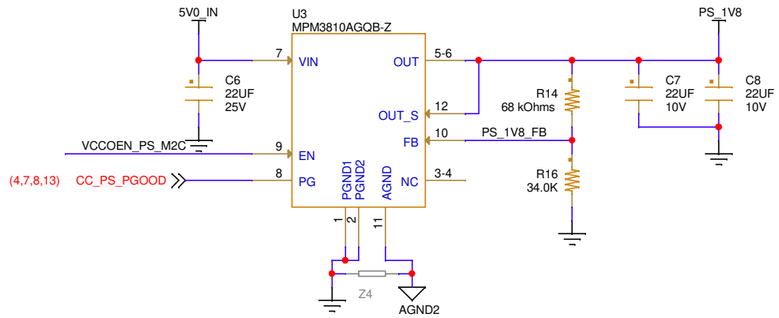
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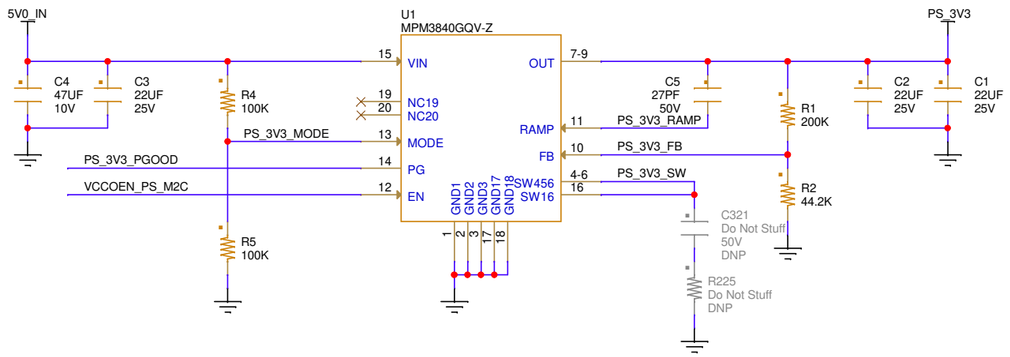
PS_1V2 REGULATOR



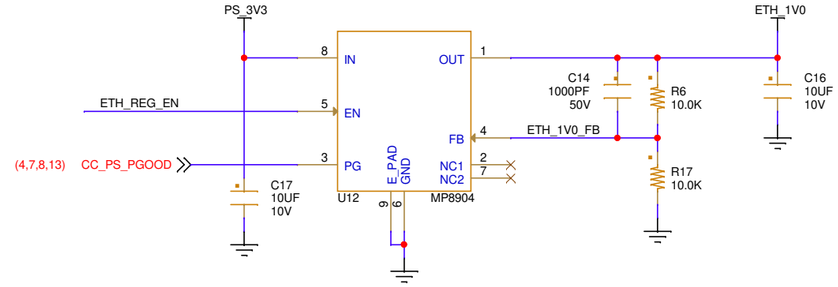
PS_1V8 REGULATOR



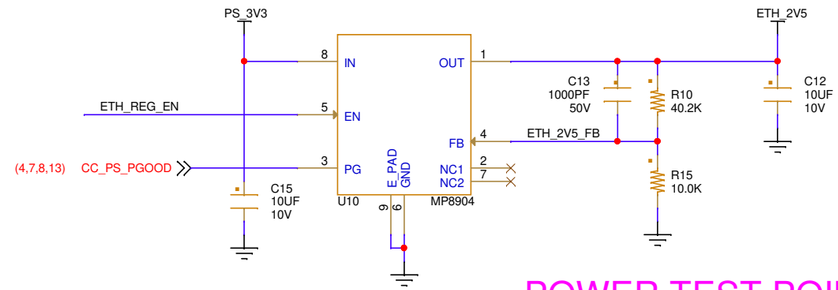
PS_3V3 REGULATOR



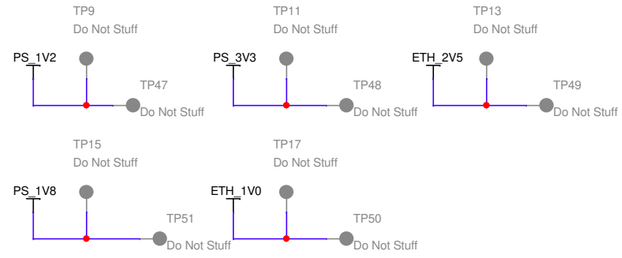
ETH_1V0 REGULATOR



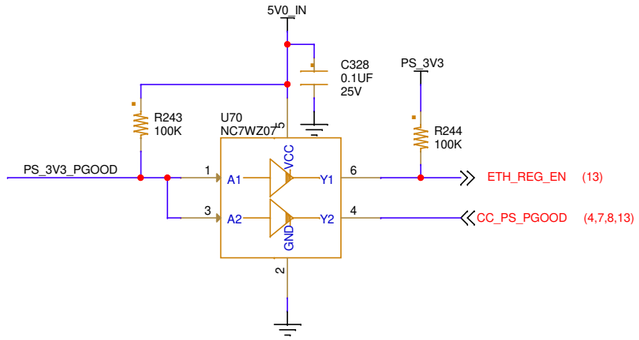
ETH_2V5 REGULATOR



POWER TEST POINTS



CAD NOTE : One TP is on Top Side and another TP is on Bottom Side of PCB



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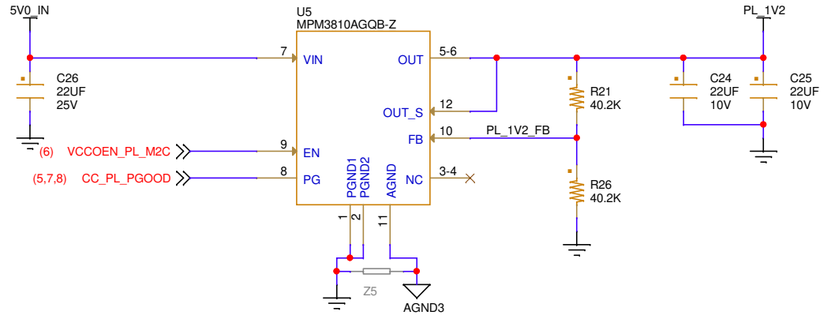
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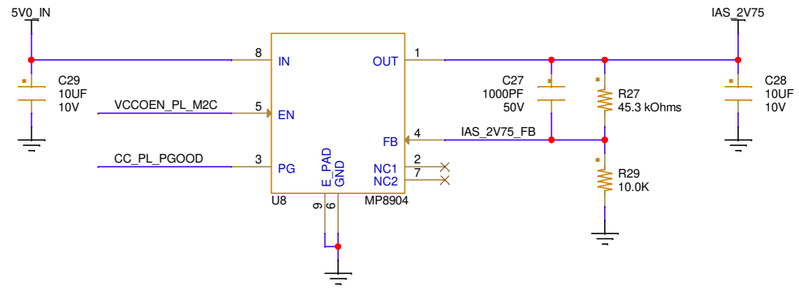
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TITLE: **SC-KV-05058**

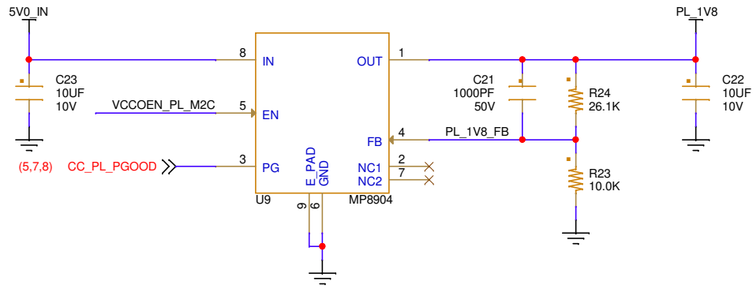
PL_1V2 REGULATOR



IAS_2V75 REGULATOR

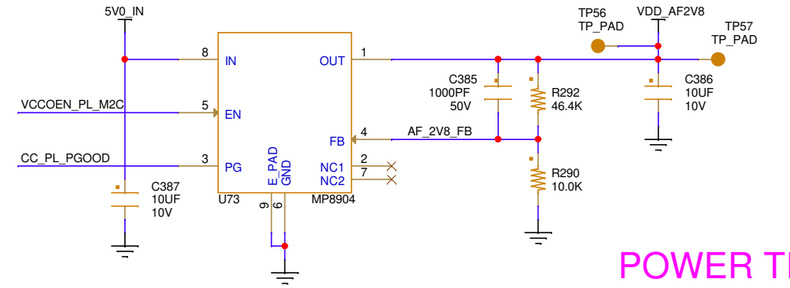


PL_1V8 REGULATOR

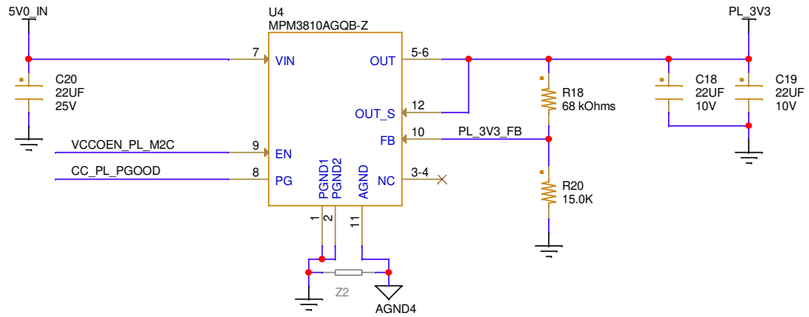


AF_2V8 REGULATOR

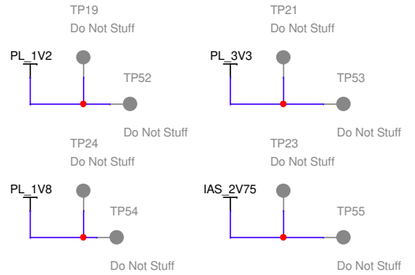
CAD NOTE : One TP is on Top Side
another TP is on Bottom Side of PCB



PL_3V3 REGULATOR



POWER TEST POINTS



CAD NOTE : One TP is on Top Side and another TP is on Bottom Side of PCB

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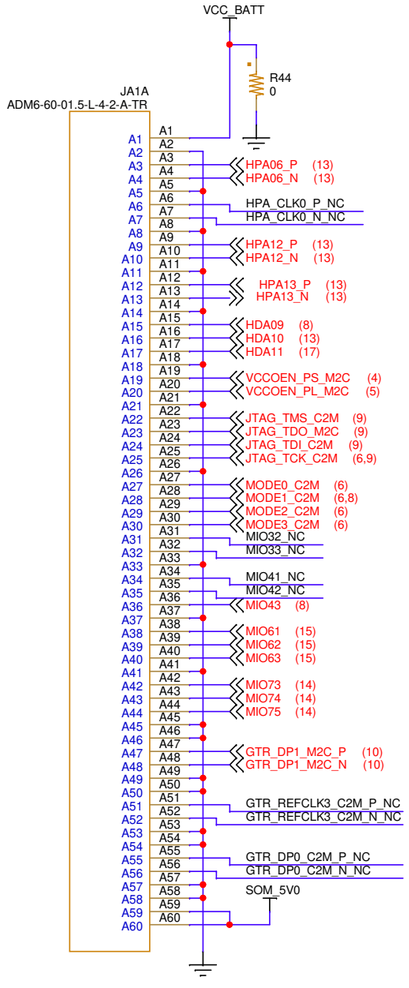
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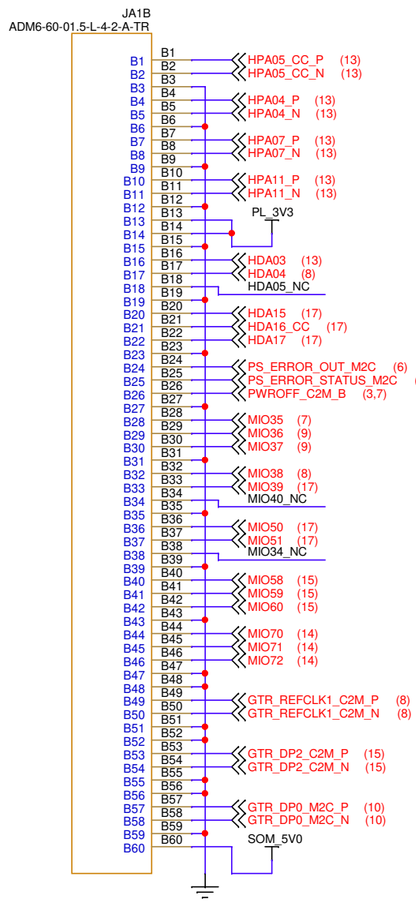
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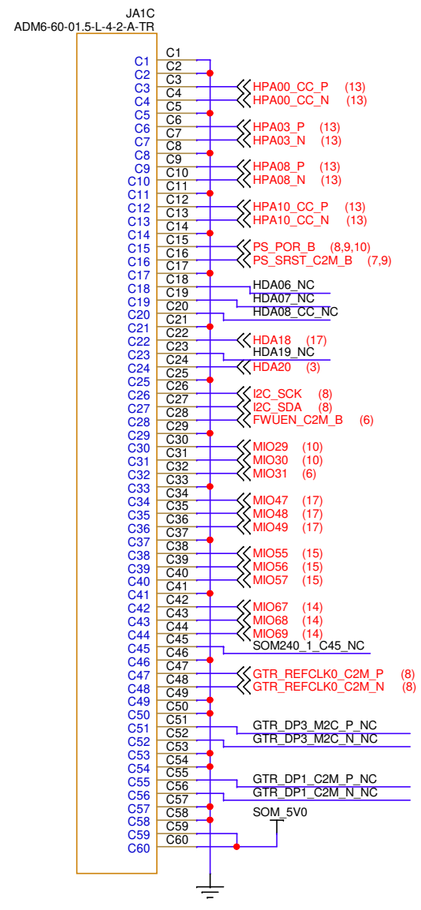
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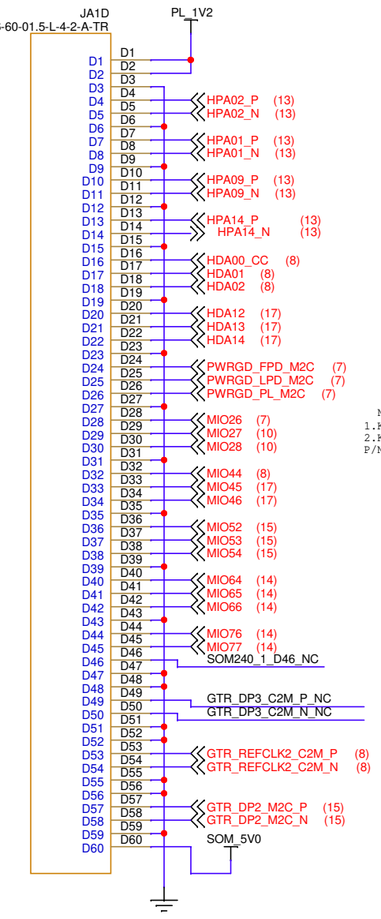
TITLE: **SC-KV-05058**

SOM240_1

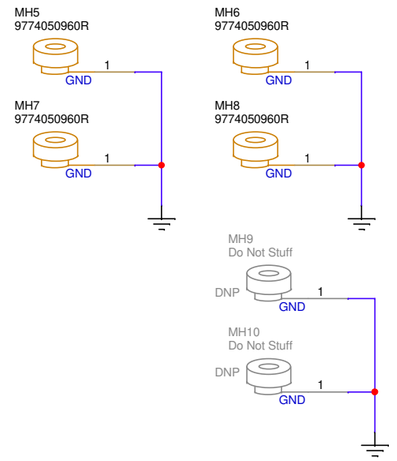






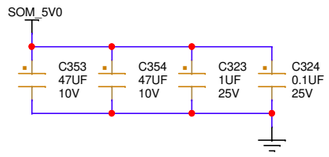


SOM STANDOFFS

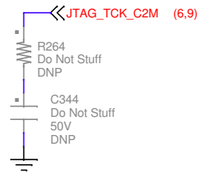


Note
1.K26 SOM + KV260 V2: Use 5mm standoff P/N: 9774050960R on MH5 ,MH6,MH7,MH8
2.K24 SOM + KV260 V2: Use 1.5mm standoff
P/N: 9774015960 (Agile PN need to be created) on MH5,MH7,MH9,MH10

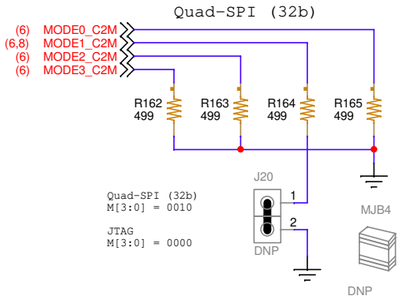
SOM_5V0 Connector Decoupling



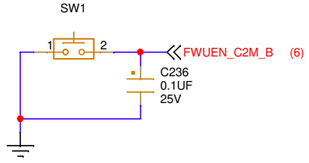
TCK TERM



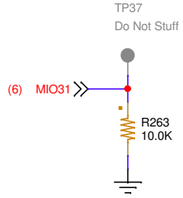
MODE PINS



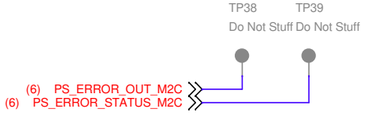
FWUEN BUTTON



SHUTDOWN



PS_ERROR_OUT/STATUS TEST POINTS



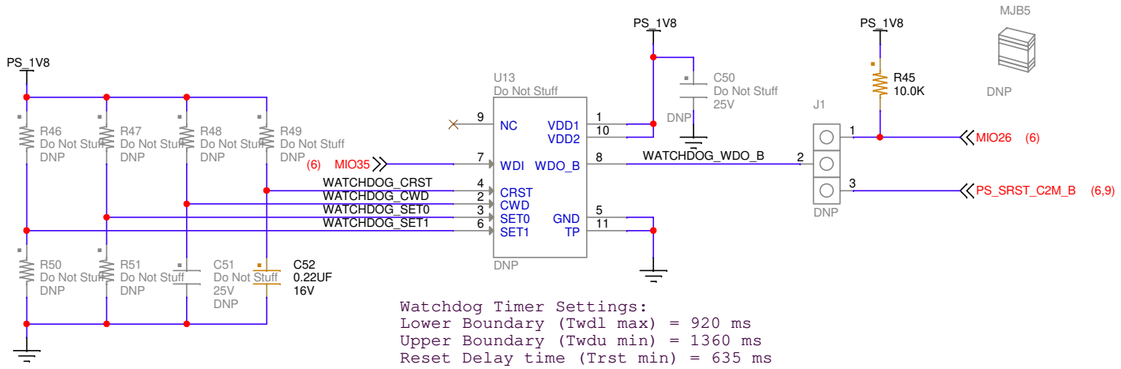
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PAGE:	**06 - SOM240_1 Connector**
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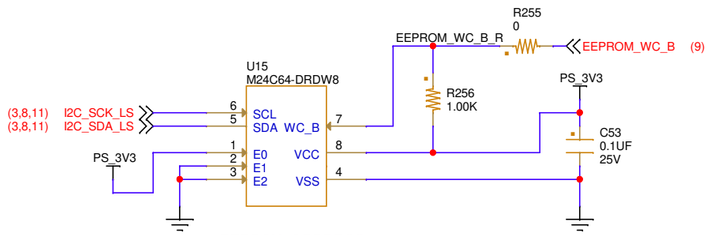
TITLE:**SC-KV-05058**

WATCHDOG TIMER



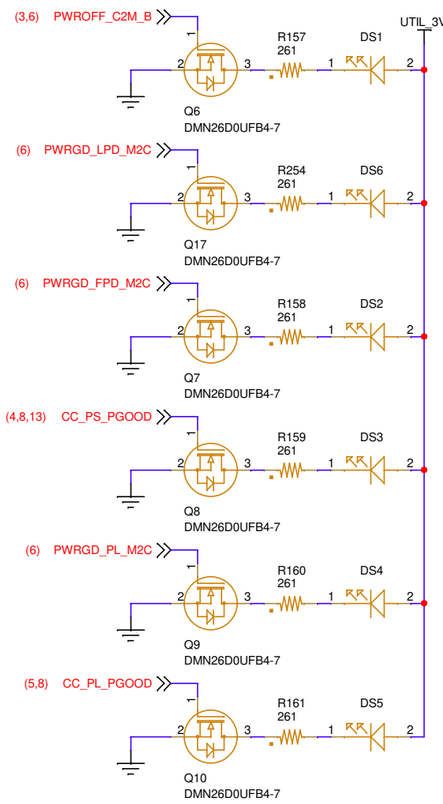

```
Watchdog Timer Settings:
Lower Boundary (Twdl max) = 920 ms
Upper Boundary (Twdu min) = 1360 ms
Reset Delay time (Trst min) = 635 ms
```

EEPROM

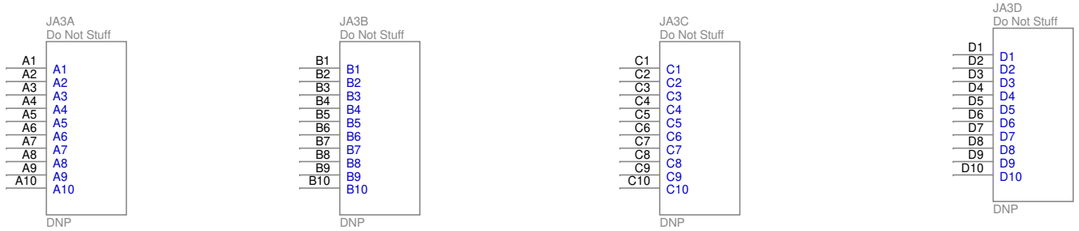



```
EEPROM ADDRESS:
0x51 and 0x59
```

POWER STATUS LEDS



K24 SOM40 CONNECTOR



NOTE- Added to consider the compatibilty with K24 SOM

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SKU: **SKIT**	

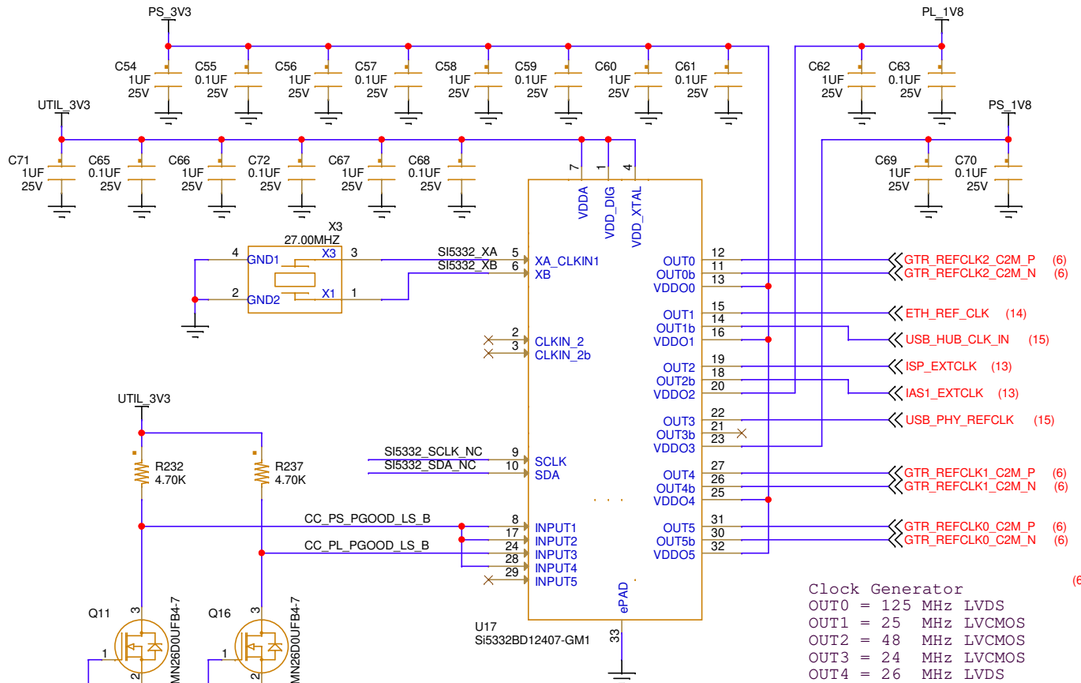
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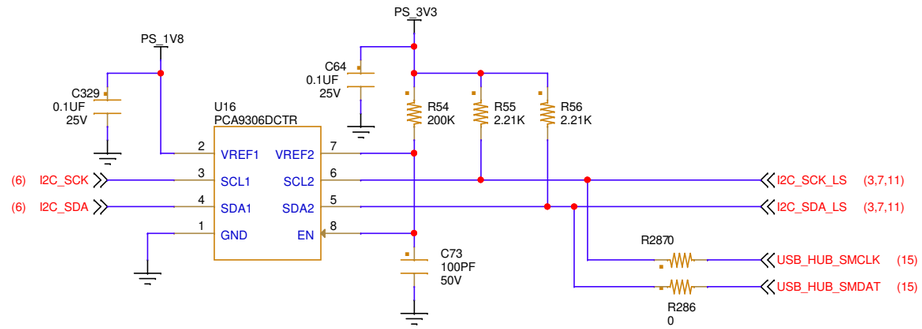
CLOCK GENERATOR



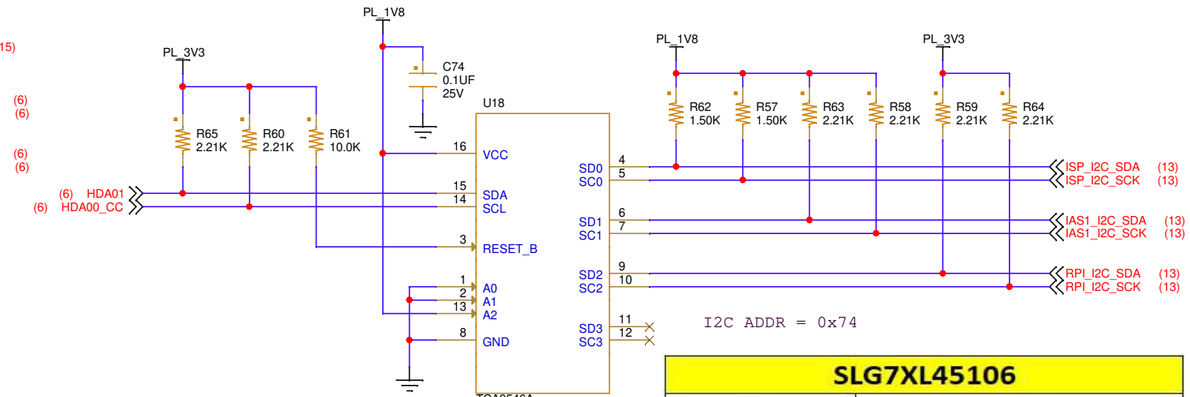
Clock Generator
OUT0 = 125 MHz LVDS
OUT1 = 25 MHz LVCMOS
OUT2 = 48 MHz LVCMOS
OUT3 = 24 MHz LVCMOS
OUT4 = 26 MHz LVDS
OUT5 = 27 MHz LVDS

Clock Generator
INPUT1 = OE_B for OUT0, OUT4, OUT5
INPUT2 = OE_B for OUT1
INPUT3 = OE_B for OUT2
INPUT4 = OE_B for OUT3
INPUT5 = UNUSED

MIO I2C LEVEL SHIFTER



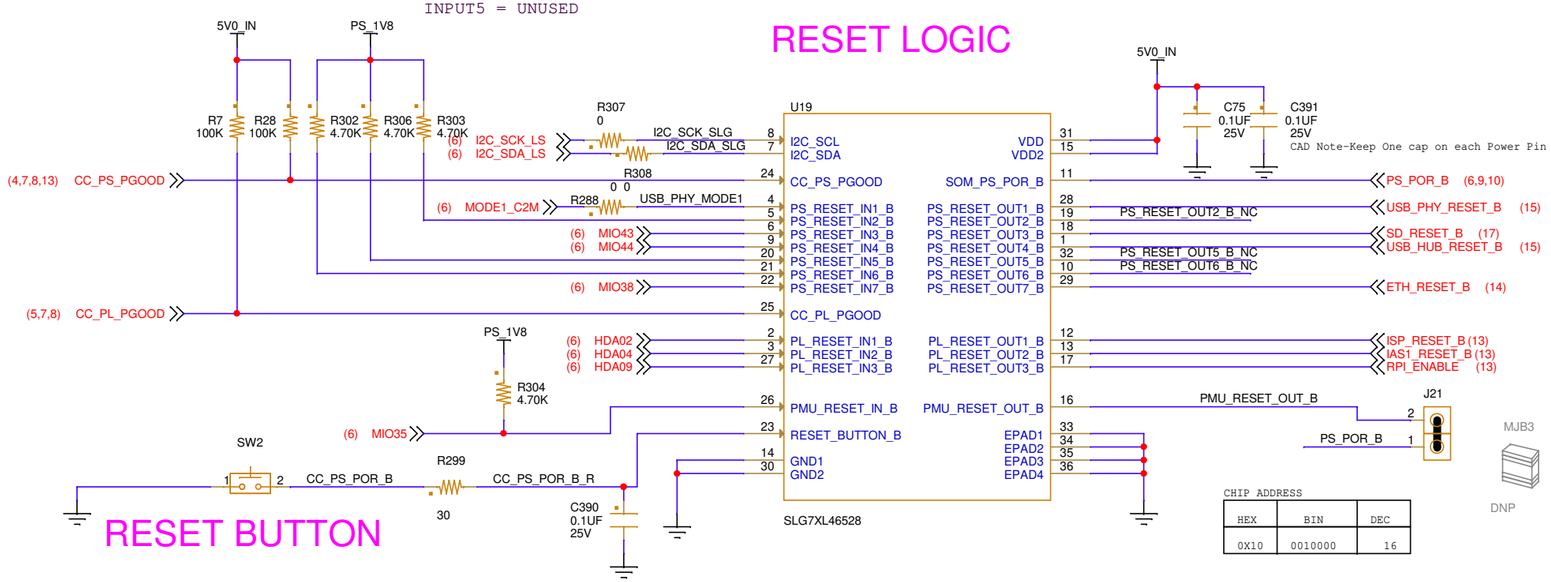
HDA I2C SWITCH



I2C ADDR = 0x74

SLG7XL45106	
IN	OUT
MODE1_C2M	USB_PHY_RESET_B
MIO43	SD_RESET_B
MIO44	USB_HUB_RESET_B
MIO38	ETH_RESET_B
HDA02	ISP_RESET_B
HDA04	IAS1_RESET_B
HDA09	RPI_ENABLE
MIO35	PMU_RESET_OUT_B

RESET LOGIC



CHIP ADDRESS

HEX	BIN	DEC
0X10	0010000	16

RESET BUTTON

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SKU:	SKIT
TITLE:	SC-KV-05058

FTDI JTAG/UART

USB_JTAG_FLTR
C78 10uF 25V
R40 10.0K
FTDI_3V3_PG
U11 MP8904
IN OUT EN FB PG NC1 NC2 E_PAD GND
USB_PWRD_VCCIO = 3.3V
C44 1000PF 50V
R36 56.2K
FTDI_3V3_FB
R41 10.0K
USB_PWRD_VCCIO
C79 10UF 10V
L2 FERRITE-600
USB_JTAG_VPHY
C82 4.7UF 6.3V
C83 0.1UF 25V
L3 FERRITE-600
USB_JTAG_VPLL
C85 4.7UF 6.3V
C84 0.1UF 25V
USB_PWRD_VCCIO
R233 10.0K
LS_OE_B
R234 10.0K
PC4_CON_PGND
Q12 DMN26D0UFB4-7
FTDI_LS_OE_B
Q13 DMN26D0UFB4-7
USB_PWRD_VCCIO
C90 0.1UF 25V
U25 SN74AVC2T245
VCCA VCCB A1 A2 B1 B2 OE_B DIR1 GND DIR2 DIR H = A-->B L = B-->A
PS_1V8
C91 0.1UF 25V
FTDI_TMS
FTDI_TDO
LS_OE_B
JTAG_TMS_C2M (6)
JTAG_TDO_M2C (6)
USB_PWRD_VCCIO
USB_PWRD_VCCIO
C80 0.1UF 25V
U22 SN74AVC4T245
VCCA VCCB 1A1 1A2 2A1 2A2 1B1 1B2 2B1 2B2 1OE_B 2OE_B 1DIR 2DIR GND GND DIR 0 B-->A DIR 1 A-->B
PS_1V8
C81 0.1UF 25V
FTDI_TCK
FTDI_TDI
JTAG_VCCO_ON
R241 10.0K
LS_OE_B
R66 0
JTAG_TCK_C2M (6)
JTAG_TDI_C2M (6)
USB_PWRD_VCCIO
USB_PWRD_VCCIO
D3 DNP
R294 100.0K
PS_POR_B (6,8,10)
USB_PWRD_VCCIO
R293 100.0K
C388 0.022uF 16V
Q24 DMN26D0UFB4-7
FTDI_PS_POR_B
Q23 DMN26D0UFB4-7
USB_JTAG_VBUS
X1 TPD4S012
VBUS NC GND D_P D_N ID
USB_JTAG_FLTR
L4 FERRITE-220
L5 FERRITE-220
J4 ZX62D_AB_5P8(30)
VBUS SHLD1 SHLD2 SHLD3 SHLD4 SHLD5 SHLD6 GND D_P D_N ID
C89 1UF 25V
C87 0.1UF 25V
C88 0.1UF 25V
R73 0
USB_JTAG_VCORE
USB_PWRD_VCCIO
U24
VCORE1 VCORE2 VCORE3 VREGOUT VCCIO1 VCCIO2 VCCIO3 VCCIO4 VREGIN
USB_JTAG_VPHY
USB_JTAG_VPLL
VPHY VPLL
USB_JTAG_DM_N
USB_JTAG_DM_P
DM DP
USB_PWRD_VCCIO
R75 1.00K
RESET#
USB_PWRD_VCCIO
R235 10.0K
ADBUS0 FTDI_TCK
ADBUS1 FTDI_TDI
ADBUS2 FTDI_TDO
ADBUS3 FTDI_TMS
ADBUS4 JTAG_VCCO_ON
ADBUS5 FTDI_LS_OE_B
ADBUS6 FTDI_PS_POR_B
ADBUS7 FTDI_PS_SRST_B
BDBUS0 FTDI_UARTB_TXD
BDBUS1 FTDI_UARTB_RXD
BDBUS2 BDBUS3 BDBUS4 BDBUS5 BDBUS6 BDBUS7
CDBUS0 FTDI_UARTC_TXD
CDBUS1 FTDI_UARTC_RXD
CDBUS2 CDBUS3 CDBUS4 CDBUS5 CDBUS6 CDBUS7
DDBUS0 FTDI_EEPROM_WC
DDBUS1 DDBUS2 DDBUS3 DDBUS4 DDBUS5 DDBUS6 DDBUS7
PWREN#
SUSPEND#
USB_PWRD_VCCIO
R268 10.0K
FTDI_PWREN_B
TP46 Do Not Stuff
FT4232HL_LQFP64
AGND GND1 GND2 GND3 GND4 GND5 GND6 GND7 GND8
USB_PWRD_VCCIO
R74 10.0K
R76 10.0K
R77 10.0K
U26 93LC56B
VCC NC ORG_NC VSS CS CLK DI DO
C94 0.1UF 25V
2.00K R78
EECS
EECLK
EEDATA
OSCI
OSCO
TEST
R79 12.0K
REF
X1 X2 X3 GND1 GND2 12.00MHZ
C103 27PF 50V
C104 27PF 50V
USB_JTAG_VCORE
C95 4.7UF 6.3V
C96 0.1UF 25V
C97 0.1UF 25V
C98 0.1UF 25V
USB_PWRD_VCCIO
C99 0.1UF 25V
C100 0.1UF 25V
C101 0.1UF 25V
C102 0.1UF 25V
C105 0.1UF 25V
EEPROM_WC_B (7)
Q19 DMN26D0UFB4-7
R262 10.0K
USB_PWRD_VCCIO
C93 0.1UF 25V
U64 SN74AVC2T245
VCCA VCCB A1 A2 B1 B2 OE_B DIR1 GND DIR2 DIR H = A-->B L = B-->A
PS_1V8
C92 0.1UF 25V
SOM_UART1_RXD
SOM_UART1_TXD
MIO37 (6)
MIO36 (6)
USB_PWRD_VCCIO
USB_PWRD_VCCIO
C325 0.1UF 25V
U65 SN74AVC2T245
VCCA VCCB A1 A2 B1 B2 OE_B DIR1 GND DIR2 DIR H = A-->B L = B-->A
PS_3V3
C326 0.1UF 25V
STDP4320_UART_RX (10)
STDP4320_UART_TX (10)
USB_PWRD_VCCIO
USB_PWRD_VCCIO
D4 DNP
R296 100.0K
PS_SRST_C2M_B (6,7)
USB_PWRD_VCCIO
R295 100.0K
C389 0.022uF 16V
Q25 DMN26D0UFB4-7
FTDI_PS_SRST_B
Q26 DMN26D0UFB4-7
PS_1V8
J3
JTAG_TMS_C2M
JTAG_TCK_C2M
JTAG_TDO_M2C
JTAG_TDI_C2M
PS_SRST_C2M_B
PC4_CON_PGND

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TITLE: SC-KV-05058

HDMI AND DISPLAYPORT SPLITTER 1

PS_3V3
R141 DNP Do Not Stuff
R145 DNP Do Not Stuff
R146 4.70K
R151 DNP Do Not Stuff
R153 DNP Do Not Stuff
R154 DNP Do Not Stuff
STDP4320_SPI_CLK BS_2
STDP4320_SPDIF_OUT0 BS_4
STDP4320_SPDIF_OUT1 BS_6
STDP4320_HOST_IRQ_OUT (11) BS_7

C211 0.1UF
C212 0.1UF
C213 0.1UF
C214 0.1UF
(6) GTR_DP1_M2C_P
(6) GTR_DP1_M2C_N
(6) GTR_DP0_M2C_P
(6) GTR_DP0_M2C_N
STDP4320_RX0_LN0_P
STDP4320_RX0_LN0_N
STDP4320_RX0_LN1_P
STDP4320_RX0_LN1_N

AVDD33
C106 0.1UF 25V
R80 249

U27A STDP4320

D6	RX0_REXT	
A5	RX0_LN0_P	
B5	RX0_LN0_N	
C4	RX0_LN1_P	
C3	RX0_LN1_N	
B2	RX0_LN2_P	
A2	RX0_LN2_N	
B1	RX0_LN3_P	
C1	RX0_LN3_N	
A6	RX0_AUX_P	DP_AUX_C_R_P
A7	RX0_AUX_N	DP_AUX_C_R_N
H11	HDMI_RX0_DDC_SCL	
K12	HDMI_RX0_DDC_SDA	
F4	RX0_HPD	SOM_DP_HPD_LS
H3	SPI_CLK	STDP4320_SPI_CLK
J1	SPI_DI	STDP4320_SPI_DI
K2	SPI_DO	STDP4320_SPI_DO
J3	SPI_CSN	STDP4320_SPI_CS_B
H4	UART_TX	(9) STDP4320_UART_TX
J2	UART_RX	(9) STDP4320_UART_RX
N7	TCLK	STDP4320_TCLK
N8	XTAL	STDP4320_XTAL
H12	RESETN	
K1	TCLK_3V3_OUT	
L1	TCLK_1V2_OUT	
D2	TEST	

AVDD12
R81 249
C107 0.1UF 25V

L2	TX0_REXT	
P7	TX0_LN0_P	HDMI_TX0_LN0_P (12)
P6	TX0_LN0_N	HDMI_TX0_LN0_N (12)
P5	TX0_LN1_P	HDMI_TX0_LN1_P (12)
N5	TX0_LN1_N	HDMI_TX0_LN1_N (12)
M4	TX0_LN2_P	HDMI_TX0_LN2_P (12)
M3	TX0_LN2_N	HDMI_TX0_LN2_N (12)
N2	TX0_LN3_P	HDMI_TX0_LN3_P (12)
P2	TX0_LN3_N	HDMI_TX0_LN3_N (12)
N1	TX0_AUX_P	
M1	TX0_AUX_N	
D13	HDMI_TX0_DDC_SCL	HDMI_DDC_SCL_LS (12)
D14	HDMI_TX0_DDC_SDA	HDMI_DDC_SDA_LS (12)
E4	TX0_HPD	HDMI_HPD_IN_LS (12)
E2	SPDIF_OUT0	STDP4320_SPDIF_OUT0

PS_3V3
R137 4.70K
R138 4.70K

AVDD12
R84 249
C113 0.1UF 25V

M9	TX1_REXT	
M14	TX1_LN0_P	DP_TX1_LN0_P (12)
N14	TX1_LN0_N	DP_TX1_LN0_N (12)
P13	TX1_LN1_P	DP_TX1_LN1_P (12)
N13	TX1_LN1_N	DP_TX1_LN1_N (12)
M12	TX1_LN2_P	DP_TX1_LN2_P (12)
M11	TX1_LN2_N	DP_TX1_LN2_N (12)
N10	TX1_LN3_P	DP_TX1_LN3_P (12)
P10	TX1_LN3_N	DP_TX1_LN3_N (12)
P9	TX1_AUX_P	DP_TX1_AUX_P (12)
P8	TX1_AUX_N	DP_TX1_AUX_N (12)
E14	HDMI_TX1_DDC_SCL	
E11	HDMI_TX1_DDC_SDA	
D1	TX1_HPD	DP_TX1_HPD_IN (12)
F3	SPDIF_OUT1	STDP4320_SPDIF_OUT1
E3	HDMI_CEC	

PS_1V2 PS_1V2
C216 10PF 50V
C215 10PF 50V
X4 27.00MHZ
GND1 GND2 X3 X1

PS_1V8 PS_3V3
R269 Do Not Stuff DNP
Q22 DMN26D0UFB4-7
(6,8,9) PS_POR_B
R168 3.01K
C239 22PF 50V
R155 10.0K
R169 10.0K

PS_3V3
R170 2.49K
R171 1.50K
R172 49.9
R173 49.9
C110 0.1UF 25V
C111 0.1UF
C112 0.1UF
DP_AUX_P
DP_AUX_N
R83 20
R85 20
DP_AUX_C_P
DP_AUX_C_N
DP_AUX_C_R_P
DP_AUX_C_R_N

PS_3V3
R87 10.0K
R88 10.0K
C116 0.1UF 25V
U31 SST25VF020B

VDD	SCK	STDP4320_SPI_CLK
HOLD#	SI	STDP4320_SPI_DO
WP#	SO	STDP4320_SPI_DI
GND	CE#	STDP4320_SPI_CS_B

PS_1V8 PS_3V3
C108 0.1UF 25V
C109 0.1UF 25V
U28 SN74AVC2T245
(6) MIO27
(6) MIO30
SOM_DP_AUX_OUT_LS
SOM_DP_AUX_IN_LS
PS_1V8
DIR H = A-->B L = B-->A

U29 FIN1019

DE	VCC	PS_3V3
DIN	DOUT_P	DP_AUX_P
	DOUT_N	DP_AUX_N
RE_B	RIN_P	
ROUT	RIN_N	
NC1	GND	
NC2		
NC3		
NC4		

PS_1V8 PS_3V3
C114 0.1UF 25V
C115 0.1UF 25V
U30 SN74AVC2T245
(6) MIO29
(6) MIO28
SOM_DP_OE_LS
SOM_DP_HPD_LS
PS_1V8
R174 100K
DIR H = A-->B L = B-->A

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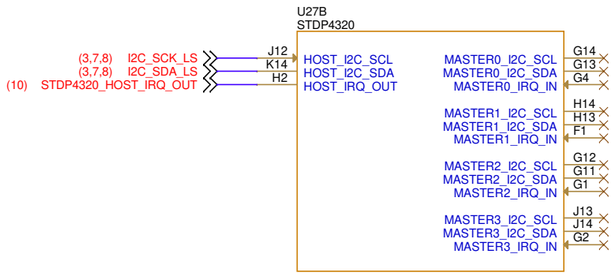
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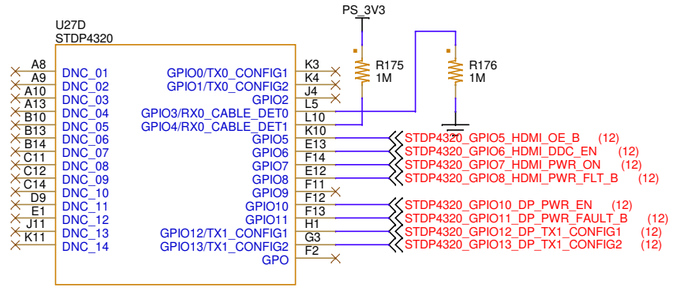
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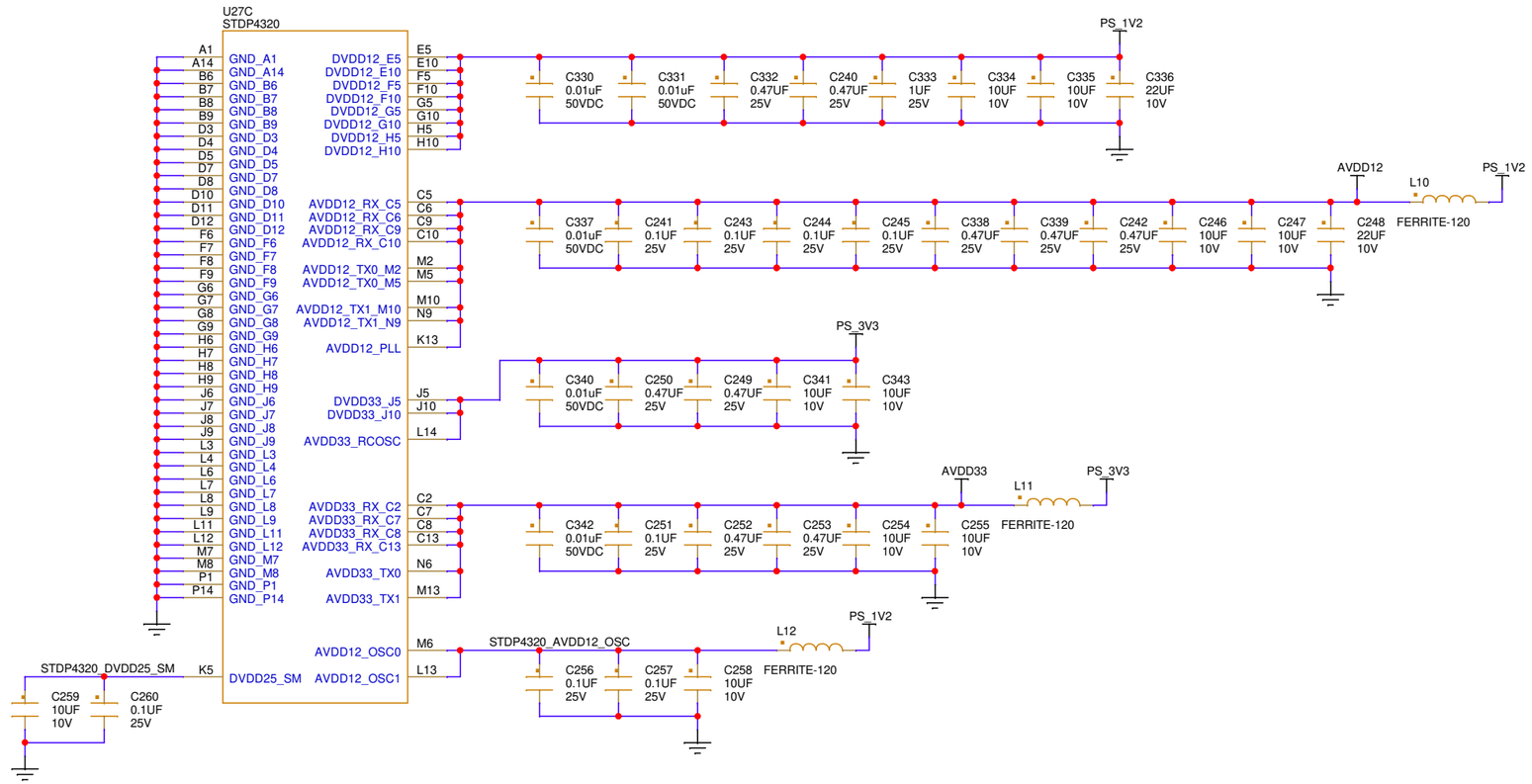
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HDMI AND DISPLAYPORT SPLITTER 2

I2C ADDR = 0x73







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HDMI AND DISPLAYPORT CONNECTORS

(11) STDP4320_GPIO5_HDMI_OE_B
(11) STDP4320_GPIO6_HDMI_DDC_EN
R185 Do Not Stuff DNP
R186 Do Not Stuff DNP
R178 10.0K
R187 10.0K
R179 10.0K
R180 10.0K
R181 10.0K
R190 10.0K
R188 10.0K
R189 10.0K
R89 4.02K

U32
SN75DP139RSBR

HDMI_OE_B 31 → OE_N
HDMI_I2C_EN 14 → I2C_EN
HDMI_DDC_EN 36 → DDC_EN
HDMI_HPDINV 38 → HPDINV
HDMI_OVS 39 → OVS
HDMI_SRC 13 → SRC
HDMI_VSADJ 15 → VSADJ
NC_1 11, NC_2 20, NC_3 40
VCC_1 3, VCC_2 8, VCC_3 12, VCC_4 19, VCC_5 23, VCC_6 28, VCC_7 32, VCC_8 37
GND_PAD 41

C217 1UF 25V
C218 1UF 25V
C219 1UF 25V
C220 1UF 25V
C358 0.01uF 50VDC
C359 0.01uF 50VDC
C357 0.01uF 50VDC
C360 0.01uF 50VDC
C221 0.1UF 25V
C225 0.1UF 25V
C222 0.01uF 50VDC
C223 0.01uF 50VDC

(10) HDMI_TX0_LN0_N — 0.1UF C226
(10) HDMI_TX0_LN0_P — 0.1UF C227
(10) HDMI_TX0_LN1_N — 0.1UF C228
(10) HDMI_TX0_LN1_P — 0.1UF C229
(10) HDMI_TX0_LN2_N — 0.1UF C230
(10) HDMI_TX0_LN2_P — 0.1UF C231
(10) HDMI_TX0_LN3_N — 0.1UF C232
(10) HDMI_TX0_LN3_P — 0.1UF C233

HDMI_TX0_LN0_C_N 2 → IN_D1+ → OUT_D1+ 29 → HDMI_TMDS_D0_N
HDMI_TX0_LN0_C_P 1 → IN_D1- → OUT_D1- 30 → HDMI_TMDS_D0_P
HDMI_TX0_LN1_C_N 5 → IN_D2+ → OUT_D2+ 26 → HDMI_TMDS_D1_N
HDMI_TX0_LN1_C_P 4 → IN_D2- → OUT_D2- 27 → HDMI_TMDS_D1_P
HDMI_TX0_LN2_C_N 7 → IN_D3+ → OUT_D3+ 24 → HDMI_TMDS_D2_N
HDMI_TX0_LN2_C_P 6 → IN_D3- → OUT_D3- 25 → HDMI_TMDS_D2_P
HDMI_TX0_LN3_C_N 10 → IN_D4+ → OUT_D4+ 21 → HDMI_TMDS_C_N
HDMI_TX0_LN3_C_P 9 → IN_D4- → OUT_D4- 22 → HDMI_TMDS_C_P

(10) HDMI_DDC_SCL_LS 18 → SCL_SOURCE → SCL_SINK 33 → HDMI_DDC_SCL
(10) HDMI_DDC_SDA_LS 17 → SDA_SOURCE → SDA_SINK 34 → HDMI_DDC_SDA
(10) HDMI_HPD_IN_LS 16 → HPD_SOURCE → HPD_SINK 35 → HDMI_HPD_IN

HDMI_5V0
R192 4.70K
R193 4.70K
R194 Do Not Stuff DNP

PS_3V3
5V0_IN
R177 10.0K
R182 Do Not Stuff DNP
R184 10.0K
C261 1UF 25V
(11) STDP4320_GPIO7_HDMI_PWR_ON
(11) STDP4320_GPIO8_HDMI_PWR_FLT_B
R183 Do Not Stuff DNP
HDMI_PWR_ON_R
HDMI_PWR_FLT_B_R

U48
TPS22948DCKR
IN 1, GND 2, ON 3, OUT 6, FLT_B 5, NC 4

HDMI_5V0
C262 0.018UF 50V
C361 10PF 50V
C263 1UF 25V

J5
HDMI_TMDS_D2_P 7 → TMDS_D0_P
HDMI_TMDS_D2_N 9 → TMDS_D0_N
HDMI_TMDS_D1_P 4 → TMDS_D1_P
HDMI_TMDS_D1_N 6 → TMDS_D1_N
HDMI_TMDS_D0_P 1 → TMDS_D2_P
HDMI_TMDS_D0_N 3 → TMDS_D2_N
HDMI_TMDS_C_P 10 → TMDS_C_P
HDMI_TMDS_C_N 12 → TMDS_C_N
DDC_VCC5V 18
MH1_GND 20, MH2_GND 21, MH3_GND 22, MH4_GND 23
HDMI_HPD_IN → HEAD_P/UTILITY 14, HEAD_N/HPD 19
HDMI_DDC_SCL → DDC_SCL 15
HDMI_DDC_SDA → DDC_SDA 16
CEC 13
TMDS_D2_SHLD 2, TMDS_D1_SHLD 5, TMDS_D0_SHLD 8, TMDS_C_SHLD 11, DDC_CEC_GND 17
C362 10PF 50V
C363 10PF 50V
C364 10PF 50V
SS-53000-001

PS_3V3
C265 0.1UF 25V
R197 Do Not Stuff DNP
R198 Do Not Stuff DNP
R195 10.0K
R196 10.0K
(11) STDP4320_GPIO10_DP_PWR_EN
(11) STDP4320_GPIO11_DP_PWR_FAULT_B
DP_PWR_EN 5
DP_PWR_FAULT_B 4

U33
STMPS2151STR
IN 3, EN 4, FAULT_B 5, OUT 1, GND 2

DP_3V3
C266 47UF 10V

(10,12) DP_TX1_HPD_IN
R199 1M

J6
0472720001

DP_TX1_LN0_C_P 1 → ML_LANE0_P
DP_TX1_LN0_C_N 3 → ML_LANE0_N
DP_TX1_LN1_C_P 4 → ML_LANE1_P
DP_TX1_LN1_C_N 6 → ML_LANE1_N
DP_TX1_LN2_C_P 7 → ML_LANE2_P
DP_TX1_LN2_C_N 9 → ML_LANE2_N
DP_TX1_LN3_C_P 10 → ML_LANE3_P
DP_TX1_LN3_C_N 12 → ML_LANE3_N
DP_TX1_AUX_C_P 15 → AUX_CH_P
DP_TX1_AUX_C_N 17 → AUX_CH_N
(11) STDP4320_GPIO12_DP_TX1_CONFIG1 13 → CONFIG1
(11) STDP4320_GPIO13_DP_TX1_CONFIG2 14 → CONFIG2
DP_PWR RETURN 20, 19
HOTPLUG_DET 18
GND 2, 5, 8, 11, 16
SHLD 21, 22, 23, 24

(10) DP_TX1_LN0_P — 0.1UF C117
(10) DP_TX1_LN0_N — 0.1UF C118
(10) DP_TX1_LN1_P — 0.1UF C119
(10) DP_TX1_LN1_N — 0.1UF C120
(10) DP_TX1_LN2_P — 0.1UF C121
(10) DP_TX1_LN2_N — 0.1UF C122
(10) DP_TX1_LN3_P — 0.1UF C123
(10) DP_TX1_LN3_N — 0.1UF C124
(10) DP_TX1_AUX_P — 0.1UF C268
(10) DP_TX1_AUX_N — 0.1UF C267

C366 10PF 50V
R200 0
C365 10PF 50V
R202 1M
R203 1M
R205 100K
R206 100K

U62
TPD4E001DRLR
IO1 1, IO2 2, GND 3, VCC 6, IO4 5, IO3 4

C271 0.1UF 25V
DP_TX1_HPD_IN (10,12)

U50
TPD4E05U06DQAR
HDMI_TMDS_D0_P 1 → D1_P, HDMI_TMDS_D0_N 2 → D1_N
GND 3
HDMI_TMDS_D1_P 4 → D2_P, HDMI_TMDS_D1_N 5 → D2_N
NC4 10, NC3 9, GND 8, NC2 7, NC1 6
HDMI_TMDS_D0_P 10 → HDMI_TMDS_D0_P
HDMI_TMDS_D0_N 9 → HDMI_TMDS_D0_N
HDMI_TMDS_D1_P 7 → HDMI_TMDS_D1_P
HDMI_TMDS_D1_N 6 → HDMI_TMDS_D1_N

U51
TPD4E05U06DQAR
HDMI_TMDS_D2_P 1 → D1_P, HDMI_TMDS_D2_N 2 → D1_N
GND 3
HDMI_TMDS_C_P 4 → D2_P, HDMI_TMDS_C_N 5 → D2_N
NC4 10, NC3 9, GND 8, NC2 7, NC1 6
HDMI_TMDS_D2_P 10 → HDMI_TMDS_D2_P
HDMI_TMDS_D2_N 9 → HDMI_TMDS_D2_N
HDMI_TMDS_C_P 7 → HDMI_TMDS_C_P
HDMI_TMDS_C_N 6 → HDMI_TMDS_C_N

U61
TPD8S009DSMR
DP_TX1_LN3_C_N 1 → D0_P
GND 2
DP_TX1_LN3_C_P 3 → D0_N
DP_TX1_LN2_C_N 4 → D1_P
GND 5
DP_TX1_LN2_C_P 6 → D1_N
DP_TX1_LN1_C_N 7 → D2_P
GND 8
DP_TX1_LN1_C_P 9 → D2_N
DP_TX1_LN0_C_N 10 → D3_P
GND 11
DP_TX1_LN0_C_P 12 → D3_N
VCC 15, NC 14, VCC 13
C269 0.1UF 25V
C270 0.1UF 25V

U49
TPD1E05U06DPYT
HDMI_5V0 → I/O 1, GND 2

U52
TPD1E05U06DPYT
HDMI_HPD_IN → I/O 1, GND 2

U55
TPD1E05U06DPYT
HDMI_DDC_SCL → I/O 1, GND 2

U58
TPD1E05U06DPYT
HDMI_DDC_SDA → I/O 1, GND 2

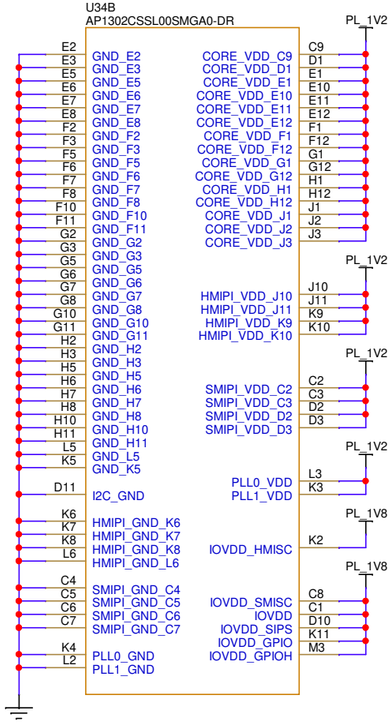
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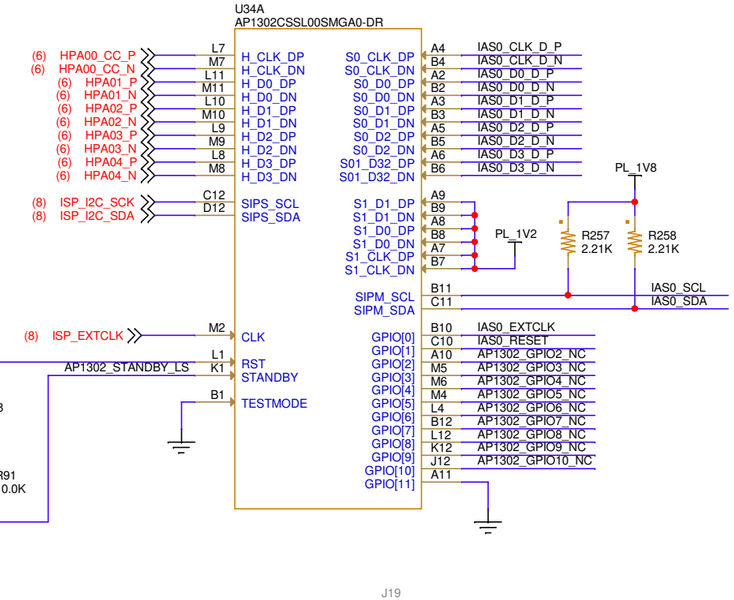
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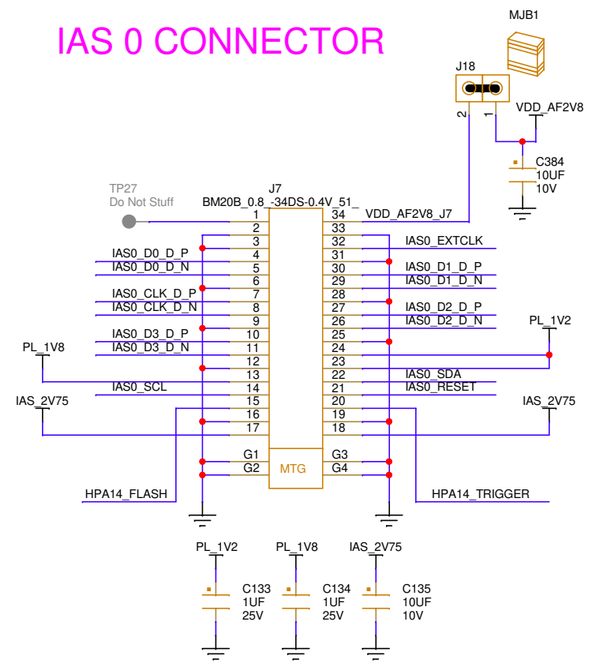
TITLE: SC-KV-05058

IMAGE CO-PROCESSOR FOR IAS 0 CONNECTOR

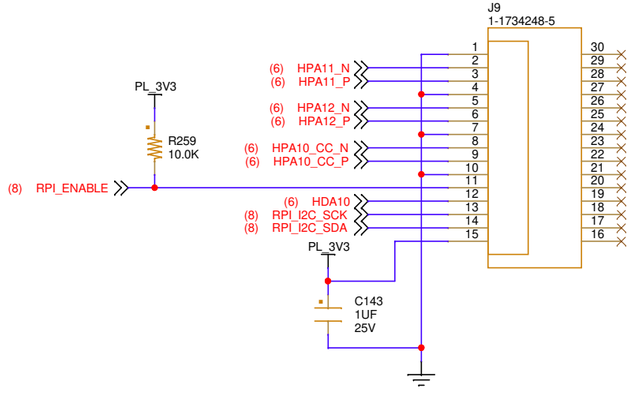




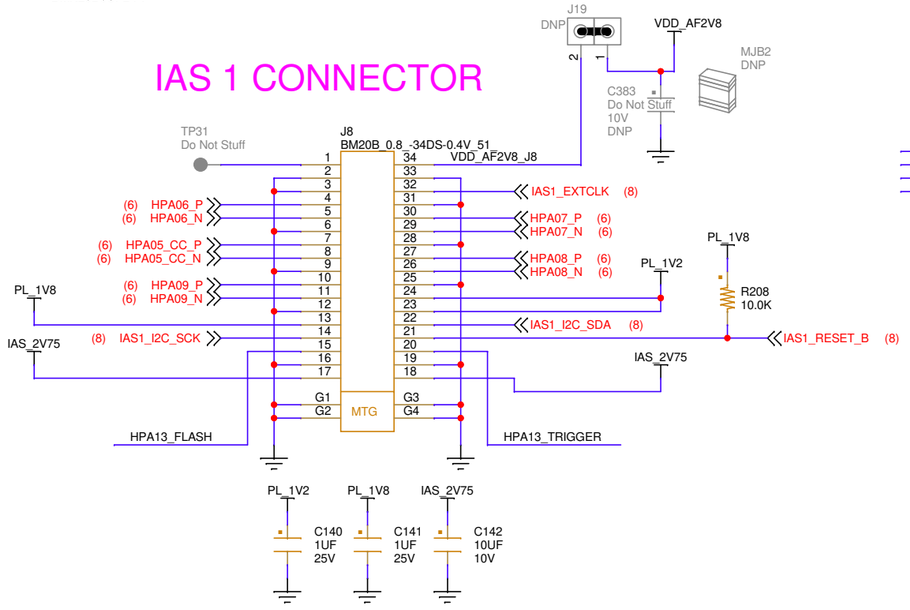
IAS 0 CONNECTOR

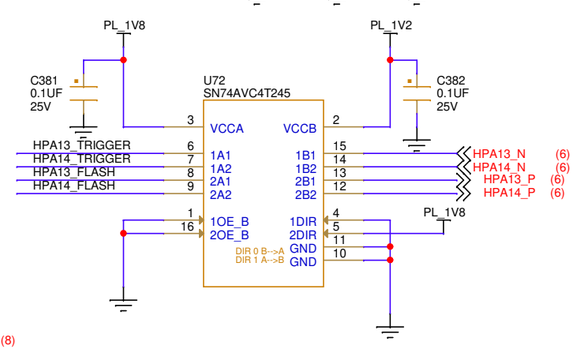


RASPBERRY PI CONNECTOR



IAS 1 CONNECTOR





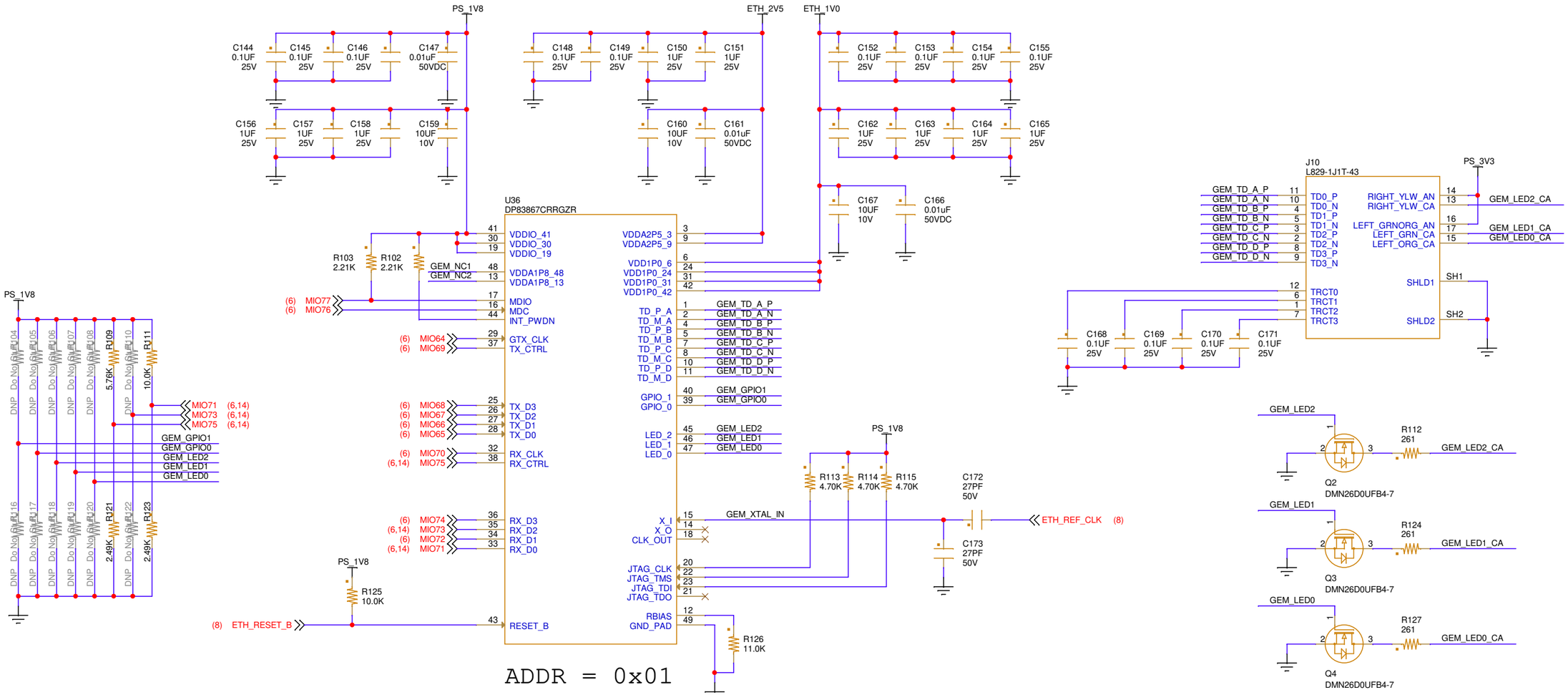
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GIGABIT ETHERNET



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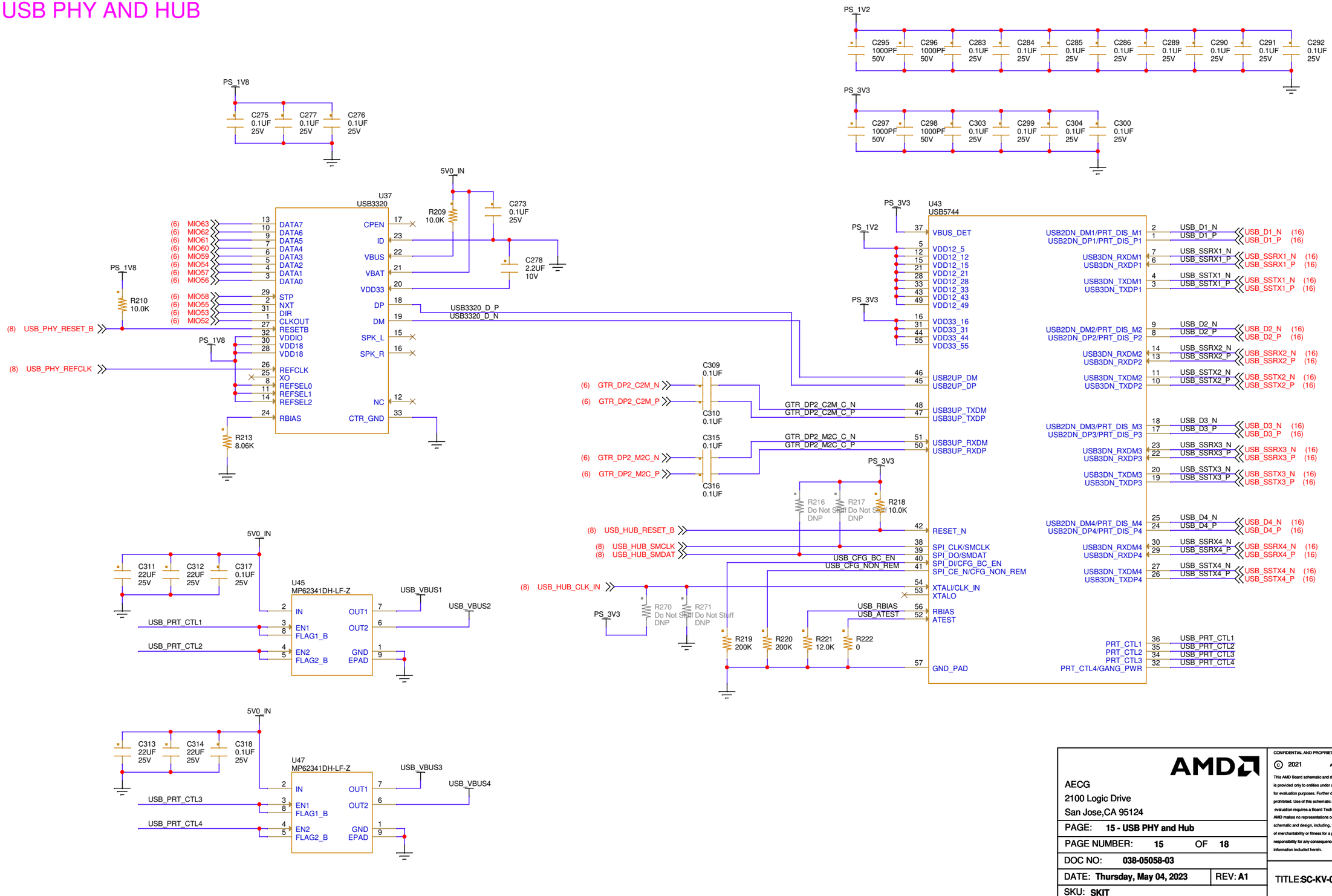
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TITLE: **SC-KV-05058**

USB PHY AND HUB



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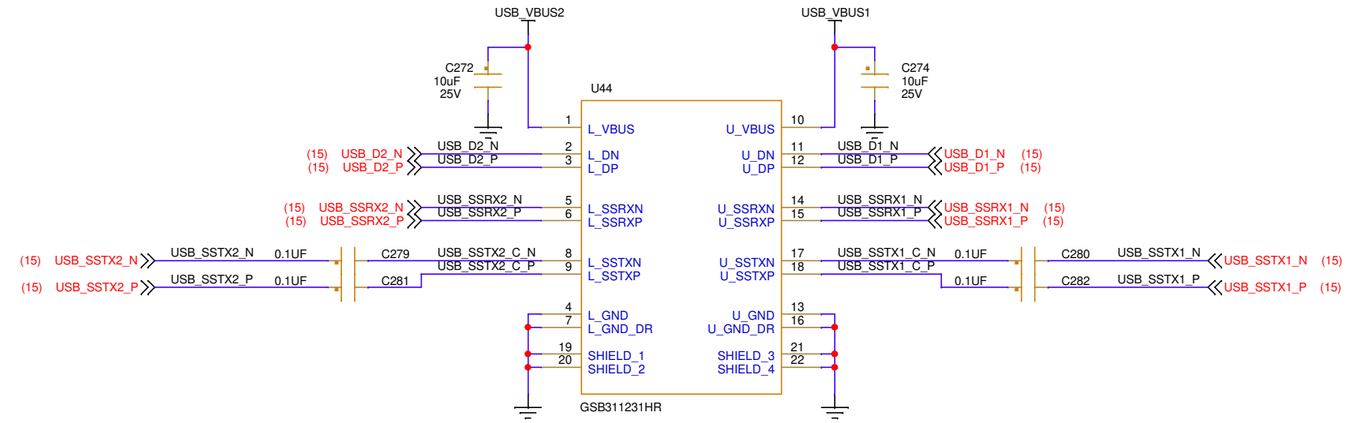
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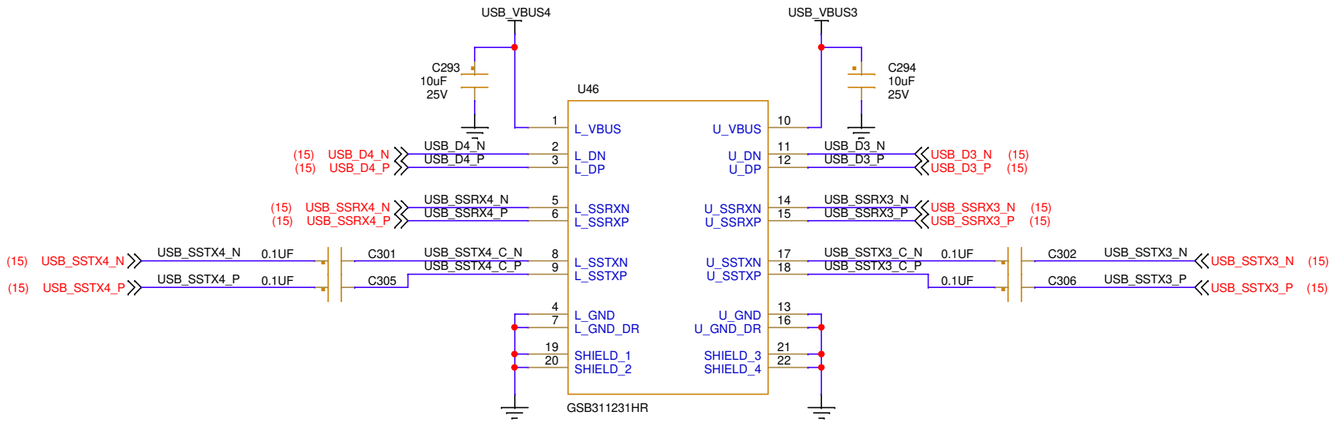
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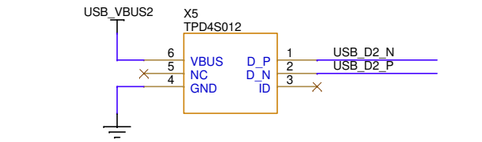
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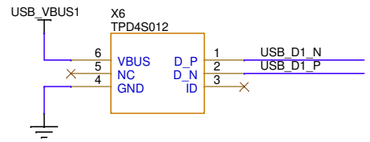
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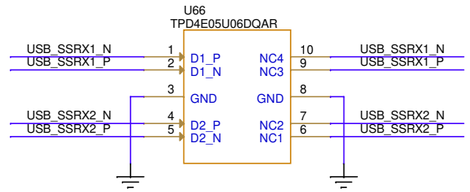
USB CONNECTORS

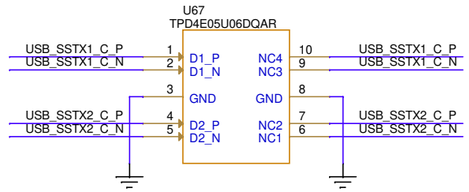


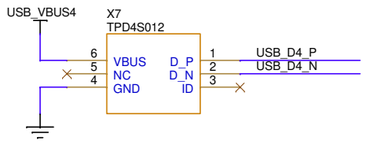


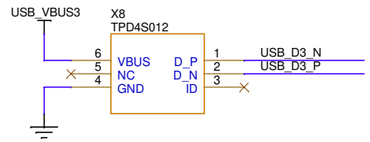


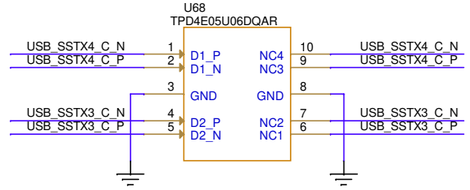


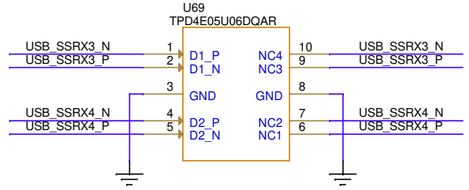












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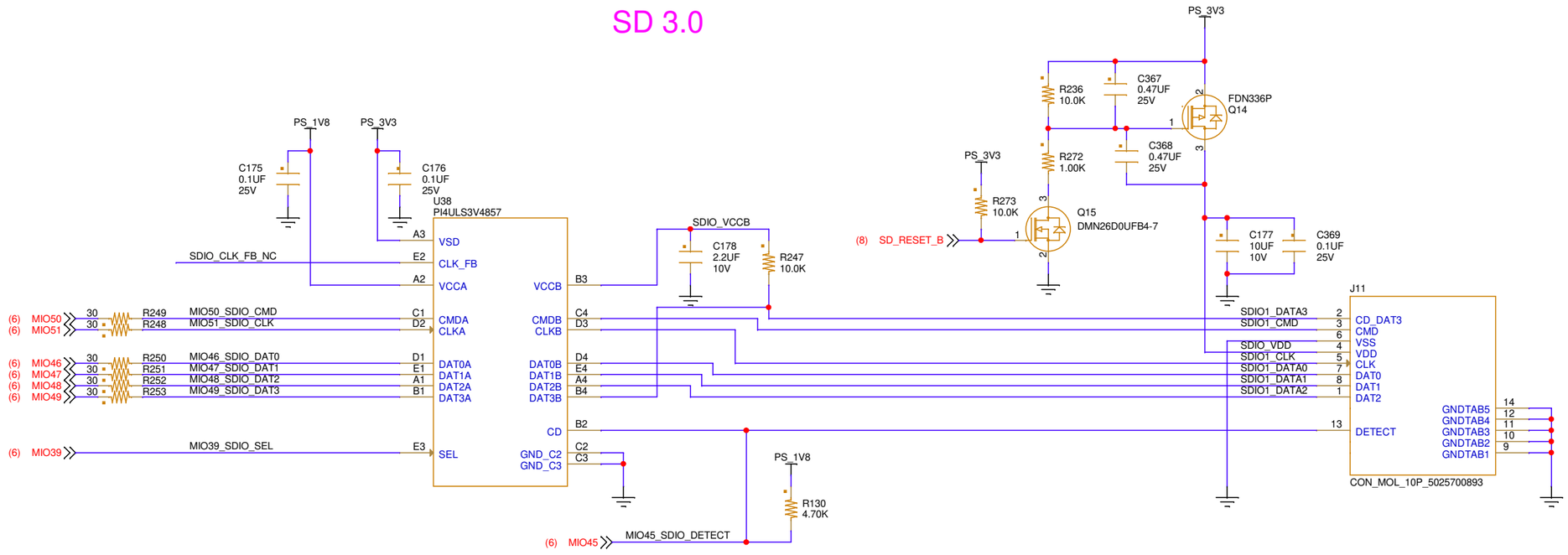
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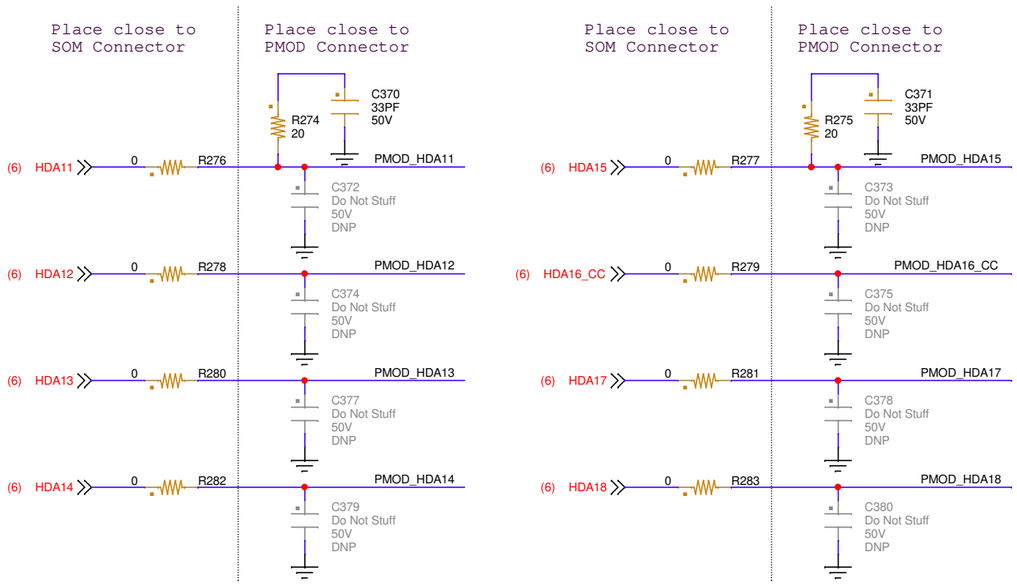
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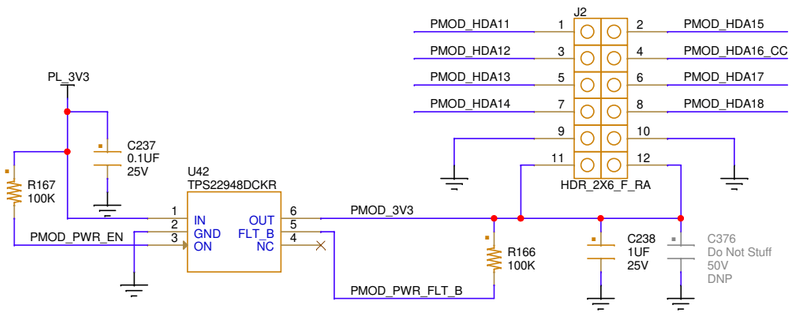
TITLE:**SC-KV-05058**

SD 3.0



PMOD





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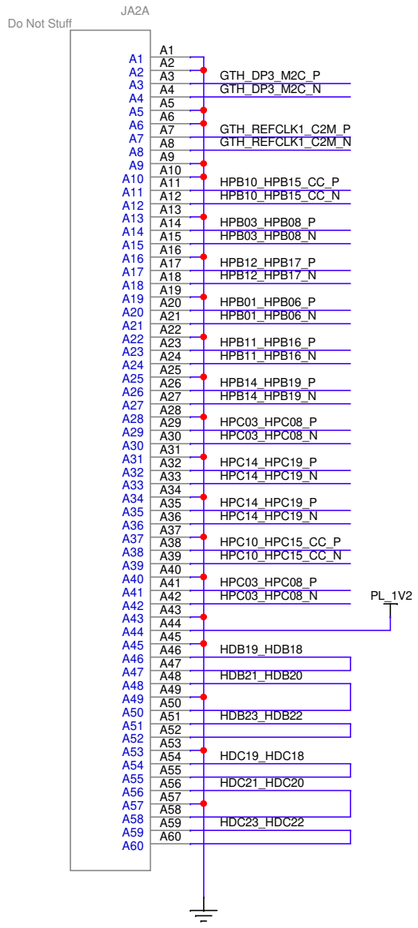
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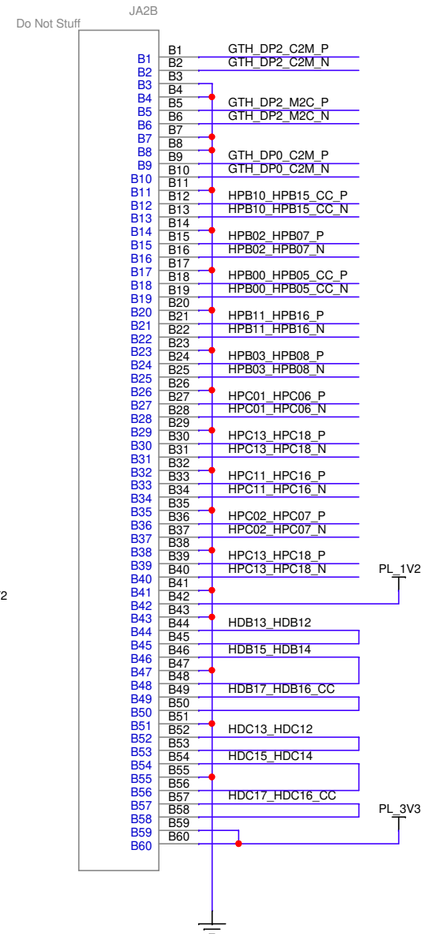
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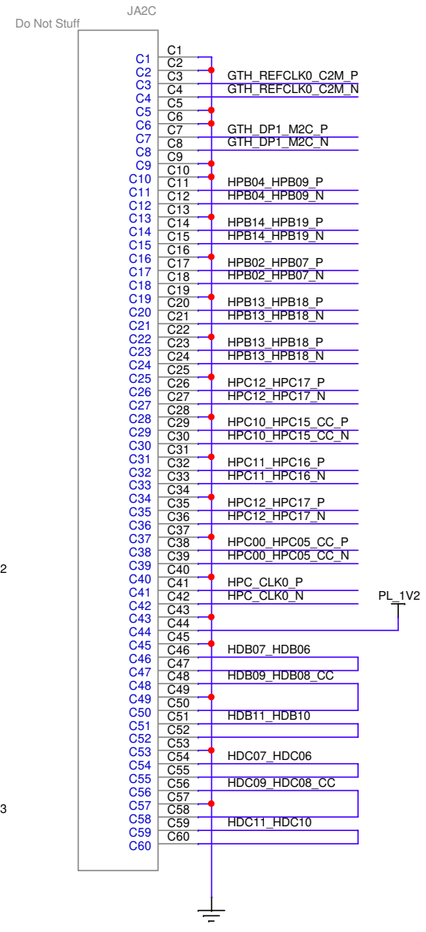
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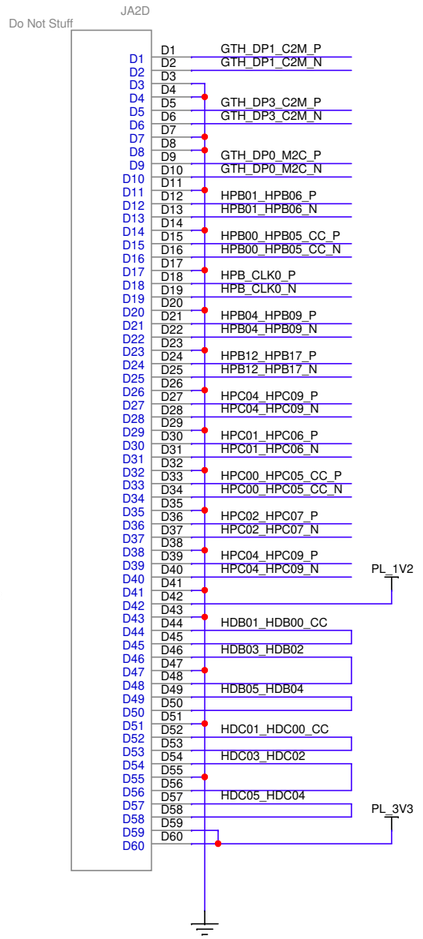
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SOM240_2 CONNECTOR

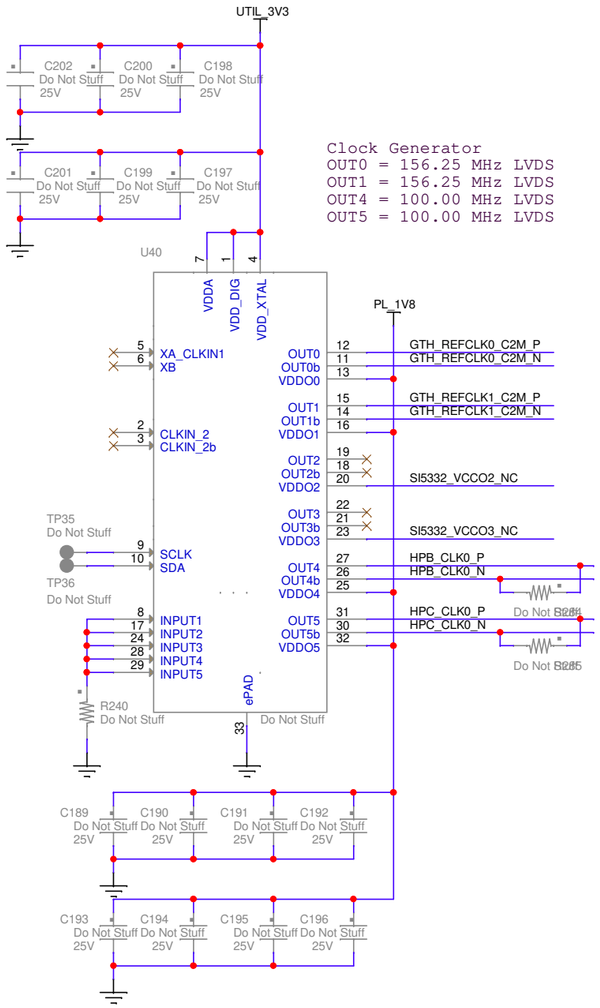




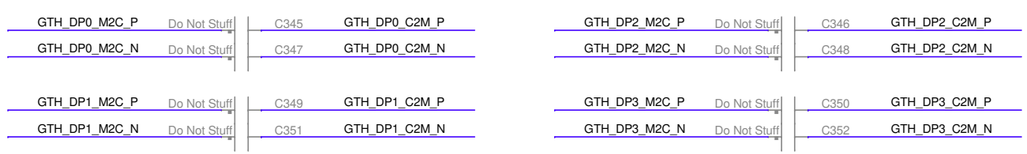




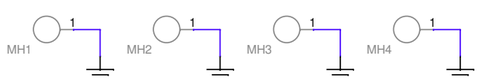
CLOCK GENERATOR FOR LOOPBACK TEST



GTH AC COUPLING



FOUR CORNER MOUNTING HOLES



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